

PYTHON PROJECT 1

TO-DO LIST MANAGEMENT SYSTEM

Industrial Training Project

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Technology: Python

Project Type: Console Based Application

Organization: DecodeLabs

INTRODUCTION

The To-Do List Management System is a simple Python application that helps users manage their daily tasks. Users can add new tasks, view all tasks, update existing tasks, delete unwanted tasks, and mark tasks as completed. The project also saves all tasks in a text file so they are available even after closing the program.

This project demonstrates the use of Python lists, functions, loops, conditional statements, file handling, and modular programming. It is designed for beginners to understand real-world Python development concepts.

OBJECTIVE

The main objectives of this project are:

To understand Python Lists.

To perform CRUD operations.

To learn File Handling.

To practice Functions.

To understand Modular Programming.

To improve problem-solving skills.

PROJECT FEATURES

- Add New Task
- View All Tasks
- Update Existing Task
- Delete Task
- Mark Task as Completed
- Save Tasks Automatically
- Load Saved Tasks
- User Friendly Menu
- Error Handling

SOFTWARE REQUIREMENTS

- Operating System: Windows 10/11
- Programming Language: Python 3.x
- IDE: Visual Studio Code

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or above
- RAM: 4GB Minimum
- Storage: 100MB Free Space

PROJECT STRUCTURE

Python Project 1/

|

└─ main.py

└─ task_manager.py

└─ file_handler.py

└─ tasks.txt

WORKING OF PROJECT

When the program starts, all previously saved tasks are loaded from tasks.txt.

The user is shown a menu.

The user selects an option.

The selected function performs the required operation.

The updated list is saved automatically.

The program repeats until the user exits.

IPO MODEL

INPUT

Task Name

User Choice

Updated Task

Task Number

PROCESS

Store task in list

Update task

Delete task

Complete task

Save task in file

Load task from file

OUTPUT

Task Added Successfully

Task Updated Successfully

Task Deleted Successfully

Completed Task

Saved Task List

```
C:\Users\Haris Awan>cd C:\Users\Haris Awan\Desktop\python project 1\py
C:\Users\Haris Awan\Desktop\python project 1\py>python main.py
Tasks loaded successfully!

===== TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit
Enter your choice (1-7): 1
Enter task title: python practice
Task 'python practice' added successfully!

===== TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit
Enter your choice (1-7): 3
Enter task ID to update: 1
```

```
Enter your choice (1-7): 3
Enter task ID to update: 1
Enter new task title: python data structre practice
Task updated successfully!
```

```
===== TO-DO LIST =====
```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit

```
Enter your choice (1-7): 4
Enter task ID to delete: 1
Task 'python data structre practice' deleted successfully!
```

```
===== TO-DO LIST =====
```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit

```
Enter your choice (1-7): 5
Enter task ID to mark as completed: 1
Invalid Task ID!
```

```
===== TO-DO LIST =====
```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit

```
Enter your choice (1-7): 1
Enter task title: test task
Task 'test task' added successfully!
```

```
===== TO-DO LIST =====
```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit

```
Enter your choice (1-7): 2
1. test task [Pending]
```

```
===== TO-DO LIST =====
```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks

```
6. Save tasks
7. Exit
Enter your choice (1-7): 5
Enter task ID to mark as completed: 1
Task marked as completed!

===== TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit
Enter your choice (1-7): 6
Tasks saved successfully!

===== TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Save Tasks
7. Exit
Enter your choice (1-7): 7
Tasks saved successfully!
Thank you for using To-Do List!
```

MODULES

Module 1

main.py

Purpose:

Controls the complete program.

Responsibilities:

Display Menu

Take User Input

Call Functions

Exit Program

Module 2

task_manager.py

Purpose:

Manages all task operations.

Responsibilities:

Module Add Task

View Task

Update Task

Delete Task

Complete Task

Module 3

file_handler.py

Purpose:

Handles file operations.

Responsibilities:

Load Tasks

Save Tasks

Read File

Write File

CONCLUSION

The To-Do List Management System is a simple and useful Python application developed to manage daily tasks efficiently. This project demonstrates the practical implementation of Python programming concepts such as lists, functions, loops, conditional statements, modular programming, and file handling. Users can add, view, update, delete, and mark tasks as completed, while all tasks are stored in a text file for future use.

Through this project, I gained practical experience in developing a menu-driven application and organizing code into multiple modules. It also improved my understanding of problem-solving, code structure, and data management. This project provides a strong foundation for developing more advanced Python applications in the future and successfully fulfills the objectives of DecodeLabs Project 1 (To-Do List).